

Agentic AI for Dynamic Factor Investing

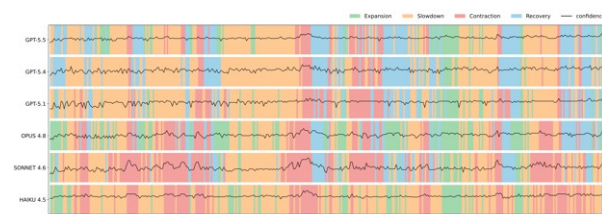
LLM driven signals for regime classification and equity factor rotation

We extend Agentic AI from regime identification into factor timing, asking whether a tightly scoped LLM agent can turn a broad macro pack into usable cycle calls. The results are encouraging, but conditional. The agent reasons coherently when real macro data are present, newer models show less directional skew, and selective committees improve the signal. The QMI (JPM's macro-navigational tool) remains the trusted systematic reference; the opportunity is to use Agentic AI as a complementary input to dynamic factor timing as model capabilities improve.

- Can an AI agent read the cycle, or is it remembering history?**
 A disciplined agent can classify regimes coherently when the prompt is well defined. We give it one job: read a broad macro pack and assign probabilities across the four common cycle phases. That coherence only matters if the agent is reading the macro *evidence*, rather than recalling *history*. Our leakage tests are reassuring: 1) removing dates, 2) levels, and even 3) regime names does not break the mapping when macro data are present. But the 4) date-only test confirms the caveat: LLMs retain historical knowledge, so out-of-sample purity cannot be guaranteed. We should not mistake confident AI output for out-of-sample insight; the model may simply be recognising known history.
- Better model capability leads to cleaner regime balance**
 Model choice matters as much as prompt design. Across six LLMs, regime calls are directionally coherent but not neutral: several models spend too much time in Slowdown and Contraction. Newer models show less skew and better alignment with the QMI reference. The multi-model committee results sharpen the point: ensembles help, but composition beats size. A selective 2-member committee of GPT 5.5 and Opus 4.8 produces the more balanced regime mix.
- From regime calls to factor rotation: do these calls make money?**
 We do not stop at classification accuracy. Using LLM-generated regimes as live rotation signals, we allocate across 4 cycle phase factor / stock portfolios: every model produces positive excess returns, and newer models generally convert better calls into better realised performance. The QMI remains the trusted systematic reference, but the results suggest Agentic AI can become a useful supplementary input to dynamic factor timing as model capabilities improve, though we suspect others will share our sigh of relief that our systematic framework remains unbeaten... for now.

Click [here](#) for an accompanying video that showcases the key points of this research.

Figure 1: Agentic AI Regime Modelling



Source: J.P. Morgan European Quantitative Strategy

See page 23 for analyst certification and important disclosures.



European Quantitative Strategy

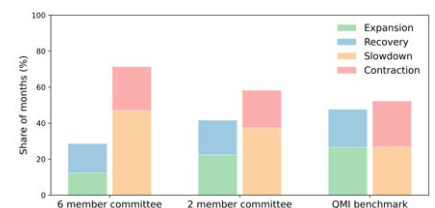
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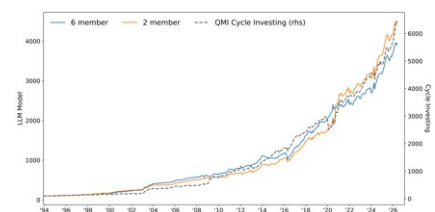
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Regime Frequency across Ensemble Agentic AI Models



Source: J.P. Morgan European Quantitative Strategy

Ensemble Agentic AI Regime Models L/S Equity Performance | Europe



Source: J.P. Morgan European Quantitative Strategy
 In this note, we extend the work of our Cross-Asset colleagues into the world of equities (see the initial paper on [Agentic AI Asset Allocation](#)).

Acknowledgments

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Introduction

What quant investors hope to achieve...

Many quant models blew up during the GFC because they were reliant on one or a few factors/ styles that became highly correlated to one another

Style timing aims to maximise returns by capturing the equity style currently in favour, and reduce drawdowns by rotating out of it before leadership changes.

Quant investors aim to generate consistent returns by removing emotion from the stock selection process. Early approaches often focused on single factors, where portfolios were built around one hypothesis at a time. For example, buying stocks at extreme valuations versus history, or backing stocks with persistent price and earnings momentum, on the view that the “tails” of the market contain the most exploitable mispricings. Despite some strong return generation with this approach, many single-factor investors have had to periodically stomach extreme drawdowns, particularly during regime shifts. The Global Financial Crisis was a stark reminder that these strategies can falter, particularly when styles become more correlated than expected.

This has pushed the industry toward multi-factor portfolios that diversify across styles so that weakness in one input can be offset by strength in others. This provides drawdown protection, however their success has led to commoditisation and ease of access, minimising the forward upward potential: marked sustained outperformance becomes harder to achieve.

Recognising that different factors can lead at different points in the cycle, the challenge shifts from identifying the single best style in the long run to managing the drawdowns that occur when leadership changes and the macro backdrop turns. Style-timing steps in to solve this trade-off issue by rotating before major drawdowns occur, without permanently diluting the upside of whichever style is in favor; although many quants remain skeptical that the timing of these turning points can be consistently predicted.

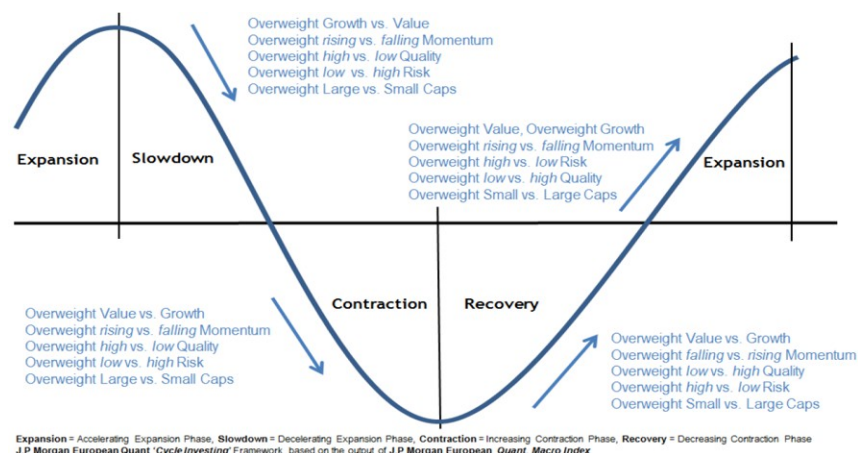
The J.P. Morgan QMI (Quant Macro Indicator) is the macro-navigational tool we built to aid style rotation. Historically, the QMI has been a suitable gauge to confirm where we are in the cycle, and more importantly has acted as a navigational tool to better understand changes in risk/reward.

In this note, we ask whether an AI agent can 1) support the same timing problem by accurately predicting upcoming economic regimes, and 2) extend this to question whether LLM predicted regimes can serve as a profitable style rotation signal.

JPM's European Cycle Investing framework links the QMI with investment style returns.

Focusing on Europe in this report, we use the J.P. Morgan European QMI.

Figure 2: JPM Quant Cycle Investing – Linking the Economic Cycle with Style Returns



Source: J.P. Morgan European Quantitative Strategy

Here we consider how an LLM may be used to diagnose the upcoming macro regime - the signal which we believe drives rotations in style leadership.

Rather than hard-coding a model that mirrors the QMI's parsimonious inputs and decision rules, we built a tightly scoped macro-style agent and gave it a single task: read a broad, generalist macro context pack and output a probability distribution over the four commonly used regimes for the upcoming month. We favour a broad pack over specialised inputs because the LLM is doing conditional reasoning. It works by weighing many pieces of evidence and updating its view as it goes. More independent "branches" (growth, labour, inflation, policy and financial conditions) give it more ways to cross-check signals and adjust the probabilities it assigns to each regime before arriving at a most likely call.

This is especially important at inflection points, when some indicators are improving while others are deteriorating. In these mixed environments, a wider set of data helps the model separate noise from a genuine regime shift. This is also where we see the benefit of newer-generation models: their stronger reasoning ability appears to make them more apt at drawing a conclusion from conflicting signals - a task that often leads even human strategists to hold diverging views! We take this as a sign that model reasoning is improving toward the level needed to form a justified view under uncertainty.

No Spoilers! Addressing look-ahead bias: when agents 'remember' the future

Look-ahead bias is a key challenge when translating any robust LLM analysis into the real-world.

A key challenge in any long-history agent backtest is hidden look-ahead bias: an LLM may recognise historical periods from the prompt even if you follow standard quant hygiene (lagging data, avoiding restatements, freezing parameters). With LLMs, the risk is not just in the data pipeline, the model itself is trained on text that describes realised economic events, so "future knowledge" can be embedded in its learned associations. For example, if we feed the model a late-2008 context pack and it assigns a high probability to contraction, it is difficult to disentangle how much of that call comes from the macro signals versus the model's prior knowledge of the Global Financial Crisis from the texts it was trained on.

Bias in LLMS is a risk to be reduced not eliminated. We test robustness across a variety of contexts ...

We therefore treat look-ahead as a risk to be reduced rather than eliminated. We anonymise the macro context packs (most notably by removing the date) and then repeat the exercise in attempt to gauge how much the model can infer from historical fingerprints versus the macro inputs themselves. Even with these controls, we cannot fully guarantee out-of-sample reproducibility, so results should be interpreted with appropriate caution.

Attenuating Models' Variation & Backtesting

Some models exhibit a pessimistic bias, selecting the Contraction and Slowdown significantly more often than the more constructive phases.

An initial read of the outputs suggested that different models may carry different directional skews in their reasoning process. We therefore evaluate the regime series estimated by the individual models, as well as a multi-model committee approach designed to dilute single-model bias, and we benchmark results against JPMorgan's QMI as a disciplined reference for factor-rotation performance.

Multi-model committees fare better than single models ...

In brief, the backtests are directionally encouraging. Across the sample, every LLM we tested generated excess returns when used as the regime signal for rotating between the four cycle phase portfolios, and results generally improve with model generation, with newer models delivering stronger outcomes than older ones. We interpret this as evidence that as models develop, better reasoning translates into more useful regime inference. Performance is further enhanced by the multi-model approach. That said, for anyone reading this as an early retirement cue, we offer a reassuring conclusion: neither any single model nor the committee approach outperformed the systematic QMI over the full sample... yet.

... though all still underperform the human-expert.

Agentic AI Regime Modelling

Prompt Design

In this first attempt to construct a simple agentic model for macro regime classification, we deliberately keep the prompt general in attempt to mirror how a macro strategist might naturally form a view, as opposed to highly engineering one, as this may embed what is effectively a systematic process and obscure what we are trying to test: the agent's ability to reason over data and form a judgment.

The agent receives a tightly-scoped role and a single task, and is then fed four fixed regime definitions and a macro data pack to reason from.

For each month, the agent is given (i) a role - a macro strategist, (ii) a clear task - produce a probability distribution across four regimes (Expansion, Slowdown, Contraction, and Recovery), and (iii) a macro context pack. The context pack is intentionally broad rather than selective, spanning a range of standard indicators across the US, UK, and Europe. We also anonymise the packs by removing the date to reduce look-ahead risk. While this does not eliminate all sources of leakage, the date is the most direct fingerprint of prominent historical events, so its removal forces the model to base its decision on the macro signals we give it, rather than on built-in knowledge.

Role

The model is put on a tight leash with strict instructions to 'Never invent data!'

We frame the agent as a senior European macro strategist embedded within a quantitative equity strategy team and give it one narrowly defined task: estimate a probability distribution across four regimes for the upcoming month. The prompt explicitly prohibits any recommendations beyond this distribution and instructs the model to remain evidence-led and never invent data. In this setup, the model's only channel of influence is the regime probabilities it outputs, with the aim of reducing the room for hallucination.

ROLE

You are a senior European macro strategist on a quantitative equity strategy team.

Your ONLY task is to estimate a probability distribution over the four phases defined below for Europe (incl UK) for the UPCOMING MONTH, using only the macro context pack provided.

Do not provide asset-class, investment, trade, or portfolio recommendations.

Communicate like a desk strategist: concise, specific, evidence-led, and never invent data!

Regime

We define the regimes in line with how we interpret them in our QMI framework, combining the level of activity against trend with the direction, i.e. whether conditions are improving or deteriorating (1st + 2nd derivative changes). This creates a fixed four-state map: Expansion, Slowdown, Contraction, and Recovery, each described in plain macro terms and paired with a brief note on expected equity leadership in that environment. Using explicit, predefined states standardises the task so outputs are comparable across dates, models, and prompt variants. We also provide a small set of interpretation rules to keep the model's reasoning disciplined.

These regime definitions reflect the idea that it is not just the level of data which affects prices (i.e. are macro conditions *good or bad?*), but the direction of movement (i.e. are they getting *better or worse?*)

In other words, where many talk in a 'first derivative language', we often find the 'second derivative' is more impactful as a guide to what is to come.

REGIME DEFINITIONS

General rule:

Regimes are defined by the level of activity vs trend (first derivative), and whether this is improving or deteriorating (second derivative). Given a point-in-time macro data context pack for a single as-of date, estimate a PROBABILITY DISTRIBUTION over all four equity regimes (non-negative, sum to 1):

Expansion

Macro/profit-cycle conditions are positive and improving. Macro signals are above trend and/or rising; cyclical risk appetite is strong; earnings revisions and equity momentum are supportive.

Typical equity winners: Value, Growth, rising Momentum, high Beta/high Risk, cyclicals, and small/mid caps. Typical laggards: high Quality, low Volatility, defensives, and large-cap bond proxies.

Slowdown

Macro/profit-cycle conditions remain positive but are decelerating. Macro signals are still supportive but falling; equity returns can remain positive, but leadership rotates toward more resilient earnings and defensive growth.

Typical winners: Momentum, Growth, Quality, large caps, defensives, and lower-risk equities. Typical laggards: high Beta/high Risk, low Quality, small caps, and deep cyclicals.

Contraction

Macro/profit-cycle conditions are negative and deteriorating. Macro signals are below trend and falling; earnings risk is elevated and investors prioritize balance-sheet strength and defensive characteristics.

Typical winners: high Quality, low Risk/low Volatility, large caps, defensives, and stable earnings stocks. Typical laggards: high Beta/high Risk, low Quality, small caps, cyclical Value, and economically sensitive equities.

Recovery

Macro/profit-cycle conditions are weak but improving. Macro signals are below trend but rising; the market begins to price a trough in earnings and rewards operating leverage and cyclical rebound exposure.

Typical winners: Value, high Beta/high Risk, small/mid caps, cyclicals, and early EPS upgrade candidates. Typical laggards: defensives, low Risk/low Volatility, and expensive Quality where earnings momentum is no longer improving.

NOTES FOR INTERPRETATION:

- Compare within-series only: interpret levels and changes relative to that series' own scale and typical variation; do not compare magnitudes across different indicators or units.

- Directionality must be interpreted per series, do not assume "up = good."

- Avoid double-counting derived flags. If the pack includes both raw series and labels, the flags are summaries of those inputs, not independent extra evidence.

- When the pack contains external series (i.e. US data) treat them as external/global conditions that may only influence Europe indirectly.

We force the model to 'show its work' by asking for rationale so we can sanity check the call is grounded in data not guesswork.

Output

We require one JSON object with a fixed structure: probabilities for the four regimes, a confidence score in [0,1], a short list of drivers, and a brief rationale. Enforcing this strict output keeps the monthly results machine-parseable, allowing us to process and backtest them.

OUTPUT FORMAT (STRICT)

Return ONLY a single valid JSON object, with no surrounding text, no markdown, and no code fences. Use this exact schema:

```
{
  "probabilities": {
    "Expansion": <float>,
    "Slowdown": <float>,
    "Contraction": <float>,
    "Recovery": <float>
  },
  "confidence": <float in [0, 1]>,
  "drivers": ["...", "..."],
  "rationale": "2-3 short sentences (<= 60 words) grounded in the provided inputs."
}
```

CONSTRAINTS

- each prob in [0,1]; all four keys present; the four SUM TO 1 (+/-0.01);
- confidence in [0,1];
- 2-5 non-empty drivers, although can be [] if no macro context is provided;
- rationale 20-400 chars, although can be "" if no macro context is provided;;
- output the JSON object only.

Where conventional model construction favours a parsimonious approach, the opposite is true of LLMs.

As explored by Dong et. Al (2024), LLMs make decisions using conditional reasoning, so more data points allow it more reference points to cross-check and reach more well informed conclusions.

Context Pack

The context pack is a compact, structured snapshot of the macro environment; providing the agent with both “level vs normal” and “direction of change” signals. A key design choice is to avoid giving the model narrowly targeted indicators that could unintentionally steer its decision-making process, and instead provide a generalist set of data¹. This spans growth (PMI), labour markets (unemployment rate), inflation (CPI/HICP vs target), policy (rates), and broader financial conditions (yield curve, real yields, USD, oil, volatility, and credit spreads). We can see what’s included in Table 1.

Each series is presented with its current level, changes over multiple horizons (1/3/6/12m depending on the series), and a rolling z-score to show how unusual the level is relative to its own history.

1. LLMs make decisions by conditioning on the examples/data points provided in the prompt (i.e., conditional reasoning), and performance depends on the demonstrations supplied. Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Jingyuan Ma, Rui Li, Heming Xia, Jingjing Xu, Zhiyong Wu, Baobao Chang, Xu Sun, Lei Li, and Zhifang Sui. 2024. A Survey on In-context Learning. In Proceedings of the 2024 *Conference on Empirical Methods in Natural Language Processing*, pages 1107–1128, Miami, Florida, USA. Association for Computational Linguistics.

Table 1: Context Pack Data

The pack includes the following series as level data, 1/3/6/12m change, and a 36m rolling Z-score

PMI	Unemployment rate	Inflation (YoY)	Policy rates	Rates	FX	Commodities	Risk	Credit
US ISM	US	US CPI	Fed funds	US 2s10s	DXY	WTI	VIX	US HY OAS
EZ Mfg PMI	EZ	EZ HICP	ECB MRO	US 10y real			MOVE	US HG OAS
UK Mfg PMI	UK	UK CPI	BoE bank rate					

Source: J.P. Morgan European Quantitative Strategy

Throughout the note we use **GPT 5.5, 5.4 and 5.1, and Claude Opus 4.8, Sonnet 4.6 and Haiku 4.5.**

All models are set to Low Reasoning mode

We fix temperature to 1.

This is less deterministic than setting it to 0, motivating multiple calls per date.

Confidence is self reported; it is part of the output we requested and gives an indication of how much conviction the model has in its call.

Model Selection

We ran the prompt described above across six **models**, using the same date-anonymised context packs throughout. The set includes three OpenAI models (GPT 5.5, GPT 5.4, GPT 5.1) and three Anthropic models (Claude Opus 4.8, Claude Sonnet 4.6, Claude Haiku 4.5). The purpose of this comparison is not to optimise any one model, but to test whether different model families or generations construct a broadly consistent macro narrative from the same information, and to identify where their regime calls diverge.

Reasoning mode can also be varied across models. “Low reasoning” produces a faster, more direct answer, and “high reasoning” allows the model to deliberate for longer before committing to a distribution. Utilising findings from our cross-asset colleagues’ work, self-reported confidence tended to be higher in low reasoning mode². We therefore ran the analysis exclusively on this setting, with the view that more decisive outputs are easier to test, compare, and stabilise across repeated runs.

As some models did not support an adjustable **temperature**, we fixed it at 1 throughout for consistency. Temperature (0–2) controls output randomness - lower values make runs more repeatable, higher values increase run-to-run variation. Fixed at 1, enough randomness remains to justify multiple calls per date and a majority-regime rule.

Figure 3 shows the most likely regime selected by each model over time, where the coloured band reflects the regime label and the black line shows the model’s reported confidence. Several patterns emerge, suggesting some cohesion in the macro narrative constructed across models. E.g. all models identify a Contraction around the 2008 financial crisis, though they differ in the time at which they first flag deteriorating conditions, with the Anthropic models tending to move into a Contraction sooner. Most models then transition into a Recovery phase, with the exception of Claude Haiku which does not classify any period as a Recovery across the full sample.

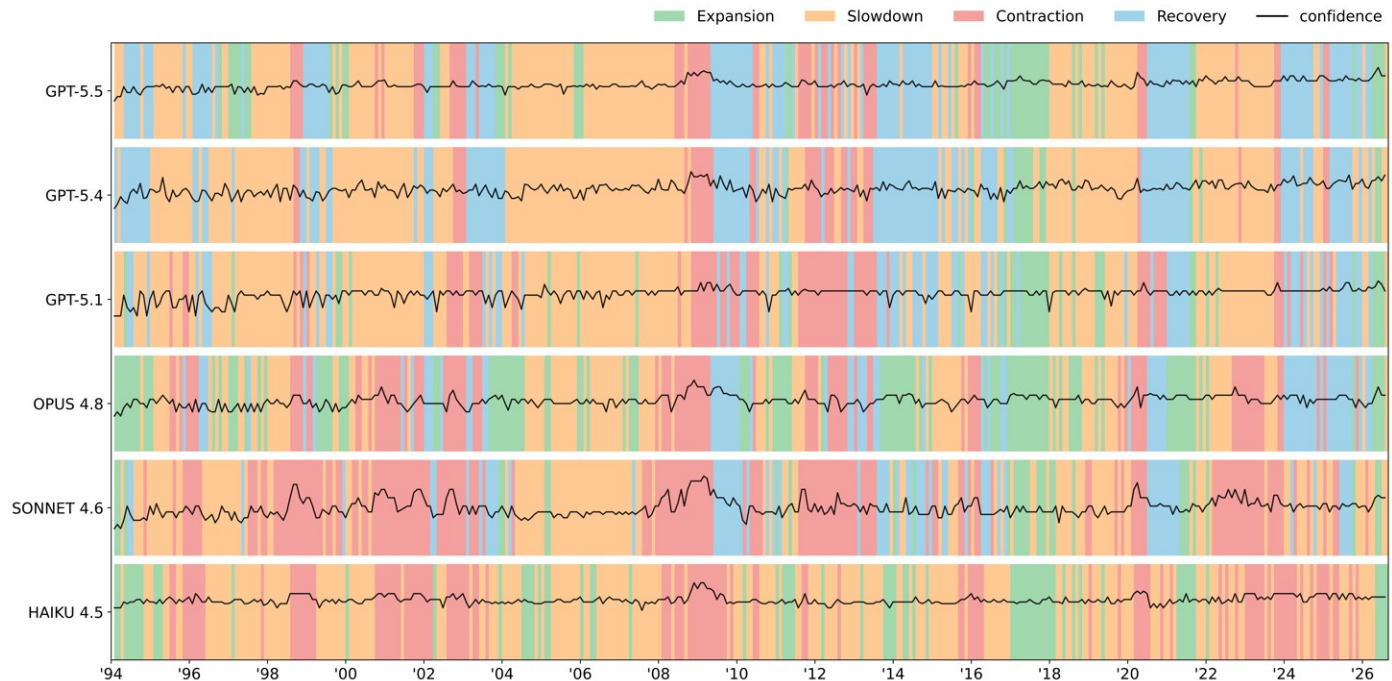
Confidence also varies through time: it tends to rise around well-known stress or inflection points (for example around 1999–2000 and 2008–2009), consistent with the idea that extreme macro environments are easier to classify than more mixed conditions.

We also observe some persistent differences in regime mix. Some models (such as Claude Haiku and Sonnet) spend much more time in Slowdown and Contraction, while others rotate more frequently between the four. This motivates an ensemble, committee aggregation approach introduced later, which intends to reduce reliance on any single model’s directional bias, and instead levers the wisdom of the crowd.

2. Self-reported confidence refers to the probability weight each model assigns to its most-likely regime. Consistent with our cross-asset colleagues’ analysis (see [Agentic AI Asset Allocation](#)), low reasoning mode concentrated more probability on the top pick, while high reasoning mode produced flatter, more hedged distributions.

Figure 3: Regime Selection by model

Models set to low reasoning mode, context pack date-anonymised



Source: J.P. Morgan European Quantitative Strategy

An ablation study aims to demystify some of the model's reasoning process.

The effect of data removal on outputs gives some insight on how that data was used.

Confidence remains intact without the date, but falls sharply without the data.

Reproducibility

We examine how regime classifications change when the anonymity of information contained in the macro context pack is varied formally known as an ablation study. This gives a sense of what information the model is using to make decisions. We test four input variants using Anthropic's frontier model (Opus 4.8):

1. **Full context:** date + macro indicators including levels, 1/3/6/12-month changes, and a 36-month rolling z-score.
2. **Date removed:** as above but with the date removed.
3. **Date & level removed:** changes + z-scores only (no date, no raw levels), intended to further reduce the risk that the model recognises historical periods from distinct absolute levels.
4. **Date only (no macro data):** date alone, aims to quantify how much regime classification can be driven purely by preexisting knowledge of economic history.

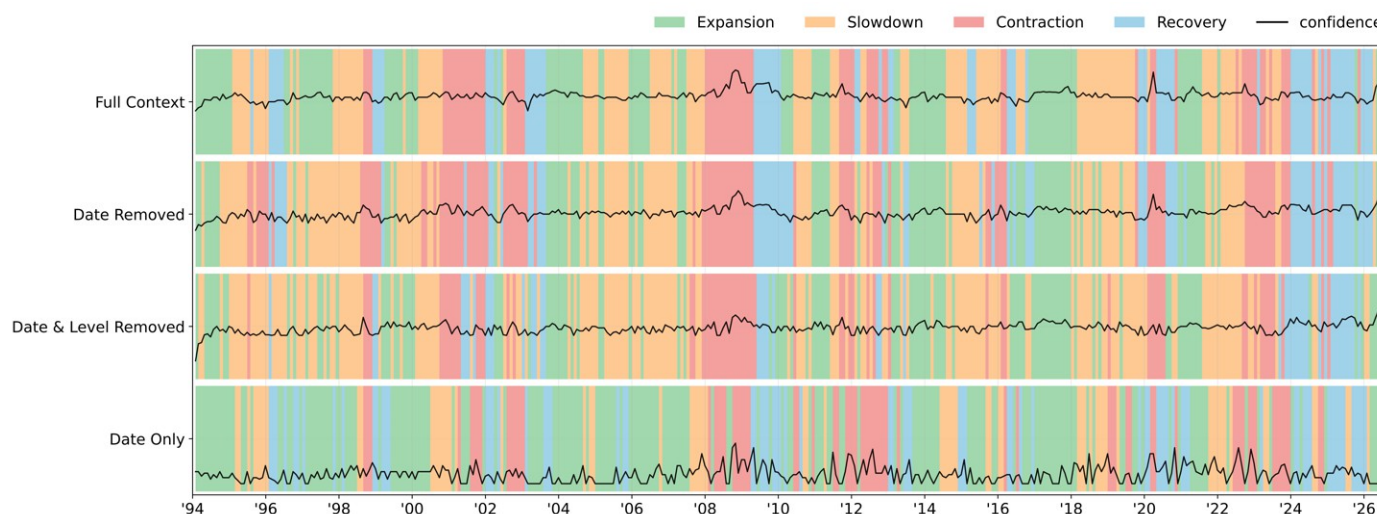
Figure 4 shows the results, which point to two conclusions. First, the model continues to produce coherent regime sequences when the date is removed, as well as when both the date and raw level data are removed. This suggests it can form a regime view from the remaining macro signals without relying solely on the date as a historical fingerprint, and it does so without a meaningful loss of confidence.

Second, the date-only test confirms why look-ahead remains a risk. Confidence falls slightly as information is progressively removed, and drops sharply when the model receives only the date. Yet the resulting regime path still resembles the macro-informed versions in some periods, especially around the Dot-Com episode, the Global Financial

Crisis and COVID, where contraction calls are made with elevated confidence. This solidifies our suspicion of the model's built-in knowledge of economic history.

The key takeaway is therefore balanced: **the model appears to rely primarily on the macro data when it is available, but enough historical recognition remains that the backtest presented later in the note should be interpreted with appropriate caution.**

Figure 4: Opus 4.8 Regime Classifications by Context Given



Source: J.P. Morgan European Quantitative Strategy

Pre-Learnt Knowledge coming to the fore: label leakage

Alongside each regime call, we asked the model to return a short rationale and a list of the drivers behind it. Inspecting these explanations, rather than the regime labels alone, offers a window into how the model reaches its conclusion, and whether it is reasoning from the macro pack or drawing on prior knowledge. The table below shows representative output from Opus 4.8 and GPT 5.5 on the date-anonymised context pack for July 2026; both return a Slowdown call for the upcoming month.

The agent is told to predict the regime for the upcoming month, so here it used July data to inform its prediction for August.

Table 2: Model Reasoning - July 2026

Model	Drivers	Rationale	Decision
GPT 5.5	- EZ and UK manufacturing PMIs remain above 50 with positive 36m z-scores - Risk conditions remain supportive: VIX calm and credit spreads tight, though spreads have edged wider - Near-term PMI momentum has softened: EZ -0.2 and UK -1.4 over 1m - EZ and UK manufacturing PMIs remain above 50, but both softened on the month - Risk conditions remain supportive: VIX calm and credit spreads tight	Europe looks positive but decelerating: PMIs are above 50 and labour markets stable, yet monthly PMI momentum has softened. Calm risk and tight spreads argue against contraction, while sticky inflation and ECB tightening cap expansion odds.	Slowdown
Opus 4.8	- EZ/UK PMIs above 50 but momentum fading (1m negative) - EZ/UK PMIs above 50 but 1m momentum softening (EZ -0.2, UK -1.4) - Tight HY/HG spreads and calm VIX signal supportive risk conditions - EZ/UK PMIs above 50 but momentum softening (1m negative) - EZ/UK PMIs above 50 but 1m momentum negative (EZ -0.2, UK -1.4)	European PMIs remain expansionary (3/3>50) with tight credit spreads and calm vol, but 1m/3m PMI deltas are rolling over and ECB is tightening into firmer inflation, tilting between Expansion and Slowdown.	Slowdown

Source: J.P. Morgan European Quantitative Strategy

The drivers are, for the most part, grounded in the pack: firm but decelerating PMIs, calm volatility, and tight credit spreads set against sticky inflation and a tightening ECB. But some phrasing gave us pause. Explanations such as *"tight spreads argue against contraction"* reason directly about the regime labels themselves, and because terms like Expansion and Contraction are well-established economic concepts, the model may be importing learnt associations between a phase name and its typical macro backdrop,

rather than inferring the regime purely from the data in front of it.

The ablation study is extended to phase names.

This raises a specific concern: is the label set itself leaking signal? To isolate this, we re-ran the exercise with the four regime names replaced by neutral placeholders (phases A, B, C and D), holding everything else fixed. If the phase names were doing hidden work, stripping them should change the calls.

Figure 5: Phase Name Anonymised Regime Classifications (Opus 4.8)

Phase A = Expansion, Phase B = Slowdown, Phase C = Contraction, and Phase D = Recovery.



Source: J.P. Morgan European Quantitative Strategy

At first glance the calls look broadly unchanged when regime names are removed: but the picture depends on **how much of the macro pack is left for the model to reason from**. Table 3 compares each month's regime call with the original labels against the call with neutral phase names (A, B, C, D), across the four context variants. We measure this two ways: raw agreement (the share of months where both produce the same regime) and Cohen's kappa, which strips out the agreement you would expect from chance alone.

Table 3: Phase Name Anonymised Agreement Comparison

Context	Agreement %	Cohen's kappa
Full Context	81%	0.74
Date Removed	83%	0.76
Date & Level Removed	80%	0.71
Date Only	59%	0.45

Source: J.P. Morgan European Quantitative Strategy

The result splits cleanly into two regimes of its own. **For the three macro-informed variants, agreement holds near 80% with kappa in a high 0.71–0.76 range: the phase names make little difference when the model has real data to work with.** In the date-only variant, however, agreement collapses to 59% and kappa falls to 0.45, a level generally read as only moderate.

Agreement is the share of months where both methods produce the same regime

Cohen's kappa adjusts this for "luck," stripping out the agreement you would expect to see if each method shuffled its regime calls randomly across months.

We find: given genuine macro signals, the agent prioritises the most informative evidence available for the task, reassuring for the validity of the approach.

Given real macro evidence the agent reasons; stripped back to just the date, it remembers

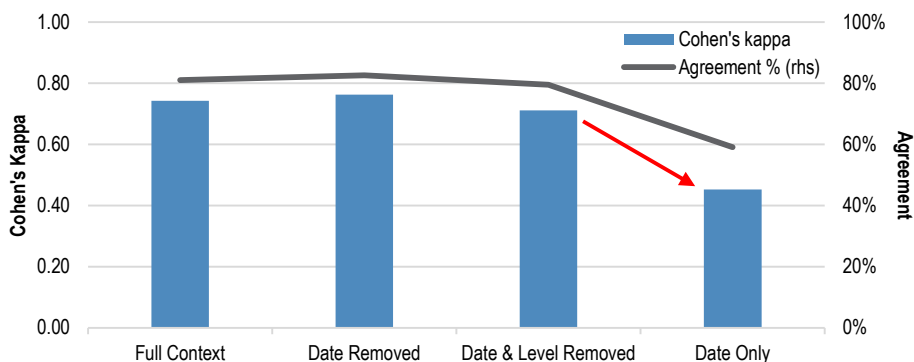
The interpretation is straightforward. **When macro inputs are stripped away, the model leans far more heavily on pre-learnt associations between the familiar regime names and the historical periods they evoke, so removing those names materially changes its calls.** When the data pack is present, the regime mapping is largely the same with or without the labels, **implying the model anchors on the evidence in front of it rather than on the names attached to it.** In short, given genuine macro signals, the agent prioritises the most informative evidence available for the task, reassuring for the validity of the approach.

A Single-Number Read on Leakage

The fall in kappa from ~0.74 to 0.45 as the data pack is stripped back is the cleanest measure of look-ahead leakage in this note. When the model is given genuine macro data, it classifies the regime primarily by reading the evidence - weighing PMIs, spreads, inflation and policy - so whether we call the states "Expansion/Slowdown/Contraction/Recovery" or "A/B/C/D" barely changes its calls (kappa ~0.74). Strip the macro data down to the date alone, and it has nothing left to read; it falls back on recognising the period: associating a familiar date and the familiar regime names attached to it, with historical episodes it learnt in training (2008 is a crisis, so Contraction). At that point the labels start doing real work, which is exactly why removing them moves the calls so much and kappa collapses to 0.45.

Figure 6: Phase Anonymisation Leakage Analysis

Regime-call agreement survives data anonymisation, but collapses when only the date remains



Source: J.P. Morgan European Quantitative Strategy

In other words, the size of the kappa drop is a direct measure of how far the model shifts from reasoning over data to retrieving from memory, and it reinforces the validity conclusion drawn from the anonymity ablation study: **given real evidence, the agent reasons; deprived of it, it remembers.**

Stability - How many calls bring confidence?

What is the Optimal N ?

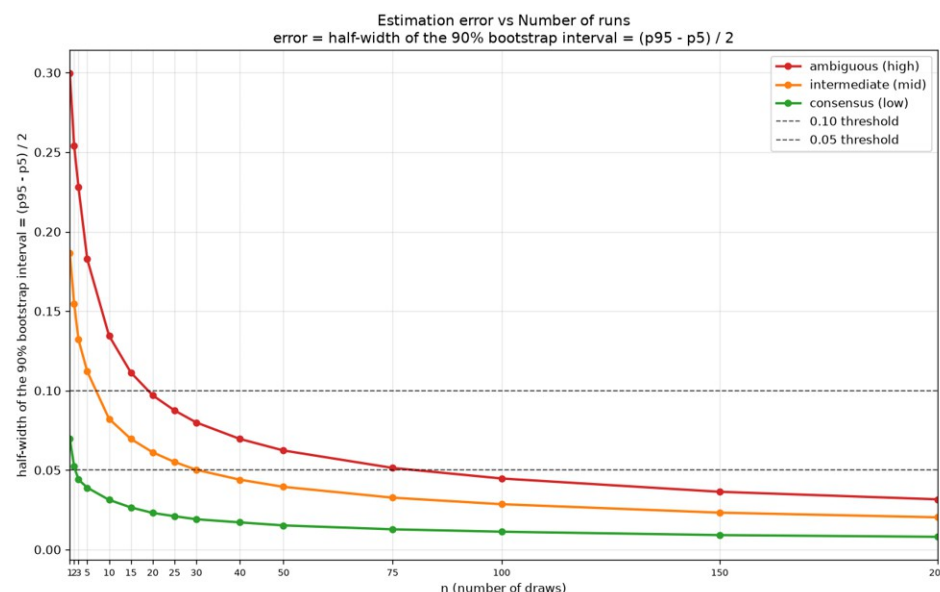
We used bootstrap resampling to pick a minimal number of runs that stabilises results but avoids the overconsumption of tokens.

As LLMs are inherently stochastic, the same model can produce different outputs to an identical prompt across runs. To improve stability and reduce the risk of effectively backtesting a one-off hallucination, we query each model multiple times per date and take the majority regime. This naturally raises a question of practicality: **what is a sufficient number of repetitions to stabilise outcomes?**

To answer this, we measure the stability of outcomes as a function of the number of repetitions, using a resampling technique known as bootstrapping. From the observed answers (200 calls per date), we repeatedly simulate what would have happened if we had run the model only n times by randomly selecting n answers from this pool (with replacement) and recomputing the majority regime. Repeating this many times gives an empirical estimate of sensitivity to run-to-run randomness at each value of n .

Figure 7: Estimation Error vs Number of Runs

error = half-width of the 90% bootstrap interval = $(p95 - p5) / 2$



Source: J.P. Morgan European Quantitative Strategy

Answer: $N=100$ is sufficient to achieve stability

Takeaway: Because LLMs are stochastic, we query each model repeatedly per date and keep the majority regime. Using bootstrap resampling, we find estimation error falls steeply with more calls before flattening, and that the cost of precision scales with how contested a date is: "consensus" dates stabilise within a few calls (e.g. dotcom, GFC), while "ambiguous" dates need close to 100 to reach the tightest error threshold. Beyond ~100 calls, gains are marginal even in the hardest cases, so **we adopt 100 runs per date as a conservative standard.**

Table 4: Minimum number of calls required for error thresholds

Category	n (<0.15)	n (<0.10)	n (<0.05)
Ambiguous (high)	10	20	100
Intermediate (mid)	3	10	40
Consensus (low)	1	1	3

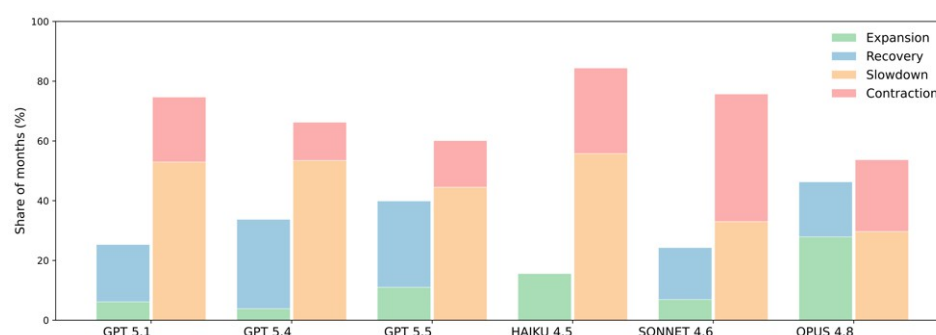
Source: J.P. Morgan European Quantitative Strategy

A Multi-Model Committee Approach

We start our analysis by assessing the regime classifications produced when we ran the prompt across the six different models over the full period (1994–2026), 100 times per month. Here we used the date-anonymised iteration of the context pack to reduce reliance on historical recognition.

At a high level, the models show a broadly similar pattern, suggesting some cohesion in the economic narrative they each construct over the last 30 years: most spend the largest share of time in Slowdown, with a smaller share in Expansion and Recovery. However, the split differs markedly model by model. For example, Claude Opus 4.8 is fairly balanced in how frequently it selects the four regimes, whereas Claude Haiku 4.5 spends the vast majority of time (over 80%) in the more negative phases of the cycle (Slowdown and Contraction), and never calls a Recovery.

Figure 8: Regime Frequency by Model (1994 - 2026)



Source: J.P. Morgan European Quantitative Strategy

These systematic differences raise a question on whether some models have an intrinsically more pessimistic (or optimistic) interpretation of the same macro information. To reduce reliance on any single model's directional bias, we therefore introduce a committee approach.

For each model and date, we average regime probabilities across the 100 runs and take the regime with the highest mean probability as that model's "call".

The committee regime is the majority vote across these six calls.

Mean probabilities are only used to break ties so that models with systematically higher self-reported confidence don't dominate the outcome.

For each month, we treat the six model outputs as independent agent views and aggregate them into a single regime call. In Figure 9 the "6 Member Committee" bar shows the regime distribution produced by this. Relative to the QMI benchmark, the 6 Member committee still appears more pessimistically skewed, with a higher share of months in Slowdown and Contraction and a lower share in Expansion and Recovery. This was expected, given 4 of the 6 models exhibit this marked negative skew, so when their outputs are pooled the aggregate distribution is pulled in that direction. We therefore also test a more selective committee. The "2 Member committee" encompasses only GPT 5.5 and Claude Opus 4.8 (the newest model each of OpenAI and Anthropic, with the regime decided by the highest average probability between them.

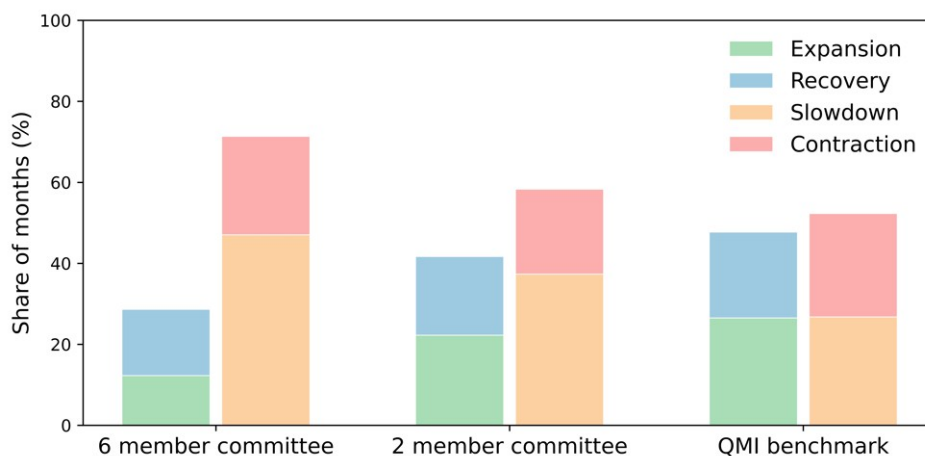
Ensemble methods and the wisdom of the crowd. The committee draws on a well-established idea in statistical learning: an ensemble of diverse models will typically outperform any single one of its members, because their independent errors tend to partly cancel when combined. Combining diverse models tends to beat any single one, as their independent errors partly cancel: but only if members are diverse *and* competent!

Aggregating the decision between all 6 models still exhibits this skew.

A selective committee comprised of the two newest models appears to be an appropriate solution.

Figure 9: Committee Approaches Regime Frequencies against the QMI

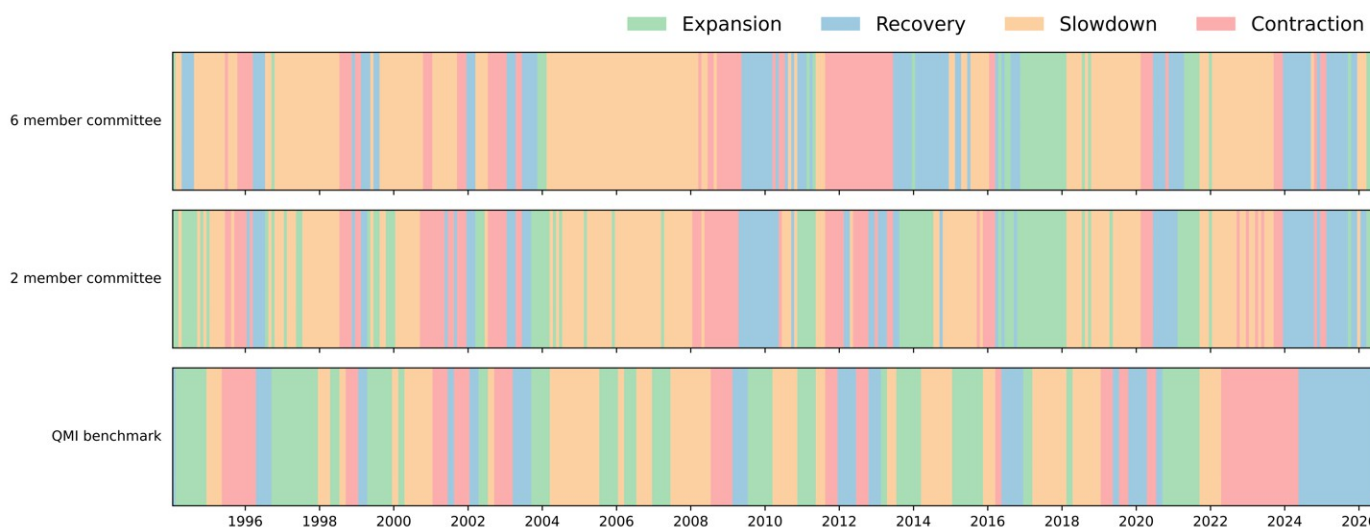
6 model committee: GPT 5.5, 5.4, 5.1, Opus 4.8, Sonnet 4.6, Haiku 4.5
 2 model committee: GPT 5.5, Opus 4.8



Source: J.P. Morgan European Quantitative Strategy

The key observation is that **the newer models produce a more balanced split across the four regimes, closer to the QMI benchmark**, rather than spending an overwhelming share of time in a single phase. Pooling six models that share a pessimistic tilt entrenches that bias rather than averaging it out. This is why a leaner committee of the two strongest, most balanced models works better: dropping the skewed members removes the majority which drags the aggregate. This is encouraging for the broader approach: it suggests newer-generation models may be better able to interpret the same macro inputs and draw more nuanced regime distinctions, which strengthens the case for agent-based frameworks as model capabilities improve.

Figure 10: Committee Regime Classifications



Source: J.P. Morgan European Quantitative Strategy

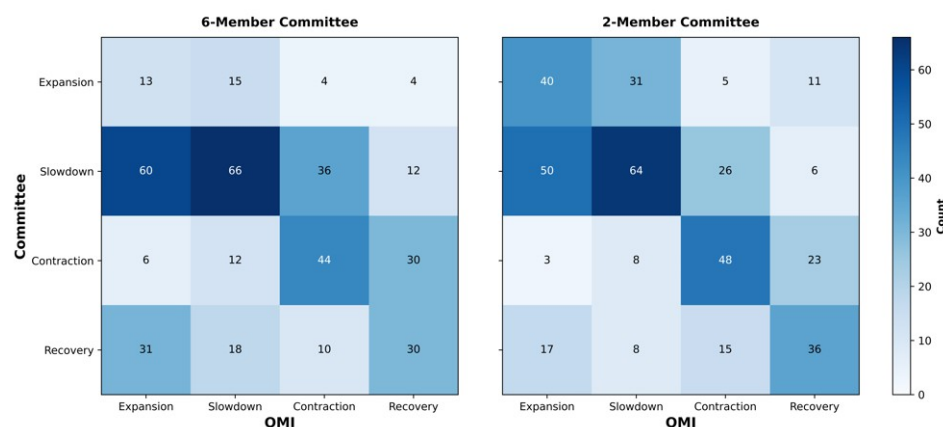
Figure 10 compares the regime series produced by the two committees. As expected, the 2-member committee tends to be a closer match to the QMI, but it also produces a more volatile path by switching phases more frequently, implying higher turnover upon implementation and therefore higher associated trading/transaction costs. The smoother path of the 6 Member committee is largely mechanical: when more independent models are being aggregated, a regime change requires a higher number of independent opinions to change around the same time, which dampens month-to-month noise.

For the following analysis we use the QMI's regime call as a metric for 'correctness', and use it to compare the accuracy of the two different committee approaches. In the confusion matrices, rows show the committee's regime call, columns show the QMI call, and each cell counts how many months that pairing occurred, with diagonals thus showing the months where the committee was correct.

The deeper shading shows more agreement between the prediction and actual QMI phase

A glance at the diagonal shows the 2 Member committee matched the QMI's regime more frequently than the 6 Member committee.

Figure 11: Committees vs QMI- Confusion Matrices



Source: J.P. Morgan European Quantitative Strategy

Both models were best at predicting a Slowdown and worst at predicting a Recovery.

The standout result is that both committees align best with the QMI in Slowdown and Contraction, which may come as little surprise given the earlier result which pointed out a broadly pessimistic skew across models. The biggest mismatch in both is calling a Slowdown whilst the QMI is in a Recovery, likely a culmination of (i) mixed macro signals around turning points making recovery difficult to diagnose and (ii) this negative tilt where LLMs spend more time than the QMI in Slowdown or Contraction. Table 5 below examines the advantages of the ensemble approach namely Precision and Recall.

Table 5: Key Accuracy Statistics

Precision: how often each committee was correct, out of all the times it called each regime (row-based accuracy)
Recall: when the QMI calls a regime, how often a committee is in the same one (column-based accuracy)

Committee	Phase	Model Count	QMI Count	Correct (N)	Precision	Recall
6-Member	Expansion	36	110	13	36.1%	11.8%
	Slowdown	174	111	66	37.9%	59.5%
	Contraction	92	94	44	47.8%	46.8%
	Recovery	89	76	30	33.7%	39.5%
	Overall	391	391	153	39.1%	39.1%
2-Member	Expansion	87	110	40	46.0%	36.4%
	Slowdown	146	111	64	43.8%	57.7%
	Contraction	82	94	48	58.5%	51.1%
	Recovery	76	76	36	47.4%	47.4%
	Overall	391	391	188	48.1%	48.1%

Source: J.P. Morgan European Quantitative Strategy

Precision = How reliable is a committee call?

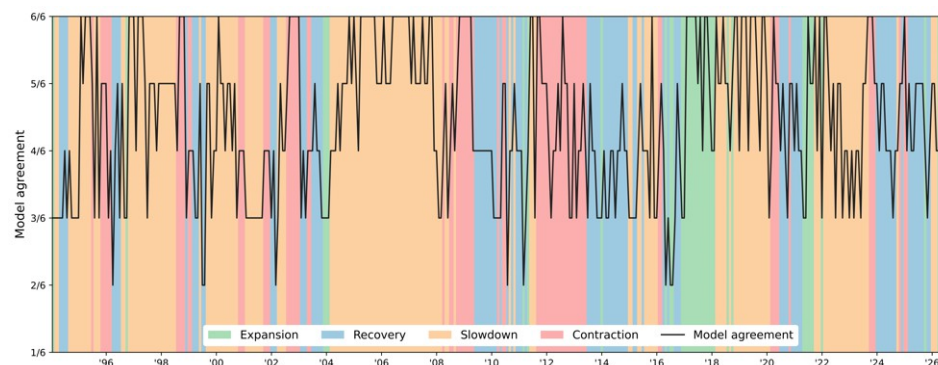
Recall = How much of the QMI does the committee capture?

The 6 Member committee only captures 11% of the Expansion calls, whereas the 2 Member captures 36%!

Overall, the 2-Member committee matched QMI more often (48.1% vs 39.1%). The improvement is most visible in Expansion (precision 46.0% vs 36.1%, and recall 36.4% vs 11.8%) and Recovery (precision/recall 47.4% vs 33.7%/39.5%), suggesting that the selective ensemble does what it was designed to do: reduce the drag from older models which struggle most with recovery and expansion calls.

The charts below show model agreement over the full period by counting how many of the models selected the same modal regime in each month. Agreement is highest around historically significant macro periods, which is unsurprising as extreme environments are easier to classify, while mixed or transitional periods leave more room for model disagreement.

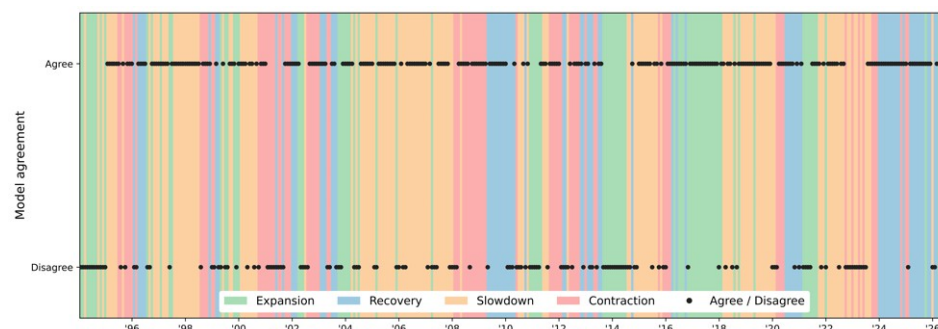
Figure 12: 6-Member Committee - Regime Classification & Model Agreement



Source: J.P. Morgan European Quantitative Strategy

- All 6 models agree in 22% of months
- 5/6 agreed in 26.9%
- 4/6 agreed in 28.1%
- 3/6 agreed in 20.7%
- 2/6 agreed in 2.3%

Figure 13: 2-Member Committee - Regime Classification & Model Agreement



Source: J.P. Morgan European Quantitative Strategy

Both models agreed in 65% of months

The remaining 35% is where the ensemble earns its keep: if the committee resolves disputed months 'correctly' more often than not, the second model adds real value.

Applied Regime Modelling and Backtesting

In effect, the LLM agents are attempting to replicate, via a probabilistic, text-based decision process, what the QMI produces through a rigorous, systematic framework, making the QMI a useful benchmark for performance.

The 2-month rule means at turning points (Expansion->Slowdown, or Contraction->Recovery) the new phase must be called in two consecutive months before allocating to its corresponding portfolio.

To gauge whether the LLM-generated regimes can serve as a useful factor-rotation signal, we ran a traditional backtest where we allocate to defined cycle phase portfolios. These multi-factor portfolios are constructed to reflect the equity styles expected to be rewarded in that regime (see Figure 2). Moreover, in the backtest, a simple 2-month confirmation rule is applied to reduce turnover cost at inflection points in the Cycle.

To keep outputs comparable across models, inference settings were standardised. All results use low reasoning mode, temperature = 1, 100 runs per date, date-anonymised context packs, and include all six models as well as both committee variants. The results below show cumulative excess and long/short returns over 1994–2026 when the LLM-selected regimes (and the committee variants) provide the monthly signal to rotate between the four cycle phase portfolios. We highlight 3 key takeaways:

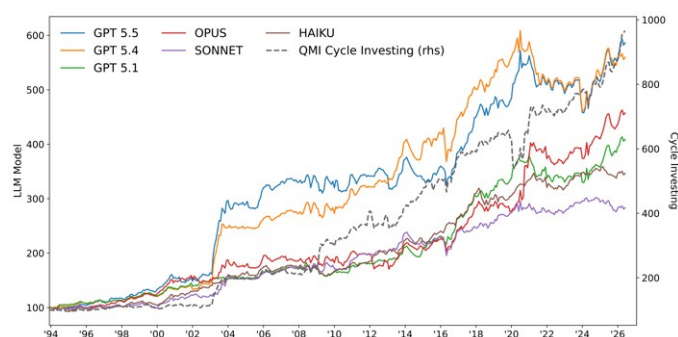
1 - A tightly scoped AI Agent can accurately and profitably detect regimes

No single LLM model matched the QMI Cycle Investing outcome over the full period, providing a sigh of relief to quants fearing imminent AI replacement, but the results are directionally encouraging: the LLM-driven strategies are broadly competitive and, in all cases, outperform the benchmark throughout the period, despite relying only on a generalist macro context pack and simple prompt.

2 - Newer model generations produce more accurate regime calls

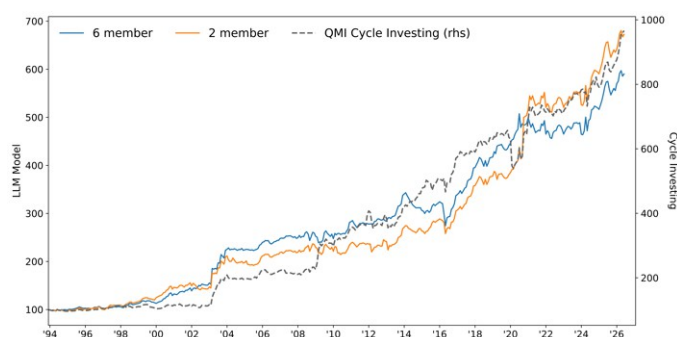
We observe a clear trend across model generations, where newer models outperform older models, suggesting they were able to predict the regime more accurately. It serves as evidence that improving model capability may make agent-based regime classification increasingly useful in future research and strategy development.

Figure 14: Agentic AI Dynamic Factor Rotation | Europe Exc. Long



Source: J.P. Morgan European Quantitative Strategy

Figure 15: Agentic Multi-Model Dynamic Factor Rotation | EU Exc. Long



Source: J.P. Morgan European Quantitative Strategy

3- Ensemble Models: Quality over Quantity when selecting the group

A further takeaway is that the 2-member committee materially outperforms any single model as well as the 6 model committee (risk-adjusted performance and drawdowns), reinforcing that ensembles can reduce directional bias but composition matters. Restricting the committee to stronger, newer models avoids weaker or systematically skewed members diluting the aggregate signal, improving results over the period.

A simple 2-member committee of the latest OpenAI model with the latest Anthropic model resulted in a strategy yielding 6% annualised excess return, with a Sharpe of 0.68, and only 10.6% max drawdown over a 30 year history, whilst this can't be guaranteed out of sample, they are not bad stats!

Table 6: Agentic AI Dynamic Factor Rotation | Exc. Long | MSCI Europe

	t-Stat	Ann. Ret	Ann. Vol	Sharpe	Hit-Rate	Skewness	Kurtosis	Max DD
GPT 5.5	3.91	5.6%	8.5%	0.66	58.5%	1.82	15.06	-20.0%
GPT 5.4	3.96	5.4%	8.1%	0.67	60.0%	2.05	18.00	-24.2%
GPT 5.1	4.25	4.4%	6.1%	0.73	59.2%	0.00	2.17	-13.7%
OPUS 4.8	3.45	4.8%	8.3%	0.58	58.5%	1.11	9.88	-16.3%
SONNET 4.6	3.03	3.3%	6.4%	0.51	57.4%	0.41	4.24	-18.0%
HAIKU 4.5	4.07	3.9%	5.6%	0.70	60.5%	0.02	1.92	-11.2%
6 member	4.21	5.6%	7.8%	0.72	60.8%	2.16	20.52	-20.1%
2 member	4.03	6.0%	8.8%	0.68	60.3%	2.26	18.54	-10.6%
QMI Cycle Investing	4.34	7.2%	9.8%	0.74	57.9%	1.44	11.36	-17.9%

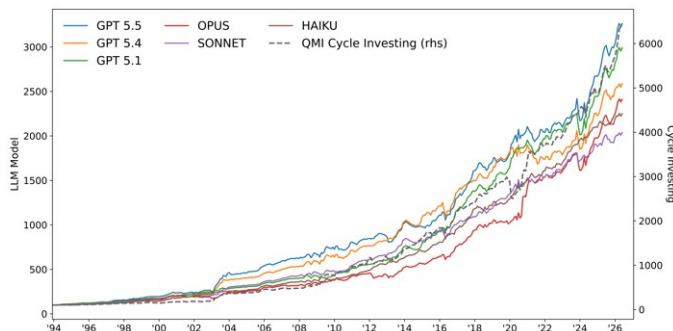
Source: J.P. Morgan European Quantitative Strategy

The key statistics are broadly competitive, particularly for the committees. The single models show a wide spread in the return–risk trade-off. GPT 5.5 and GPT 5.4 lead on return (5.6%, 5.4%) but carry large Drawdowns (-20.0%, -24.2%), which is costly in a rotation framework. Sonnet 4.6 is weakest on both Return (3.3%) and Sharpe (0.51). The 6-Member committee improves consistency, with Sharpe 0.72 and the highest Hit-Rate (60.8%), while keeping Return competitive (5.6%), but at the cost of Drawdowns. The 2-member committee is the most allocator-relevant: it delivers the highest agent return (6.0%) and the tightest drawdown (-10.6%) with a solid hit-rate (60.3%), reiterating the point that ensemble benefits are real but they come from member quality not quantity. Clearly, Cycle Investing still anchors the frontier for upside capture and drawdown control.

Despite strong results, outperformance during distinct historical periods highlights how look-ahead bias can give LLM strategies a superficial boost.

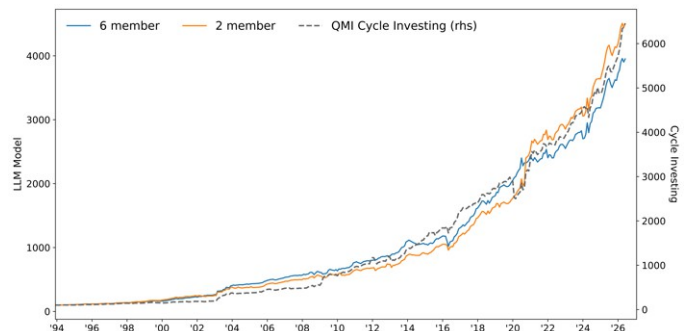
Two caveats matter: First, much of the newer models' outsized outperformance appears concentrated around 2020, so it should be treated cautiously: extreme periods are where any residual look-ahead or historical-recognition effects can be most impactful, even with date anonymisation, and it is hard to disentangle how much of the apparent success reflects pre-existing knowledge of the pandemic's macro footprint. Second, Sharpe can be misleading for older models: Haiku 4.5 and GPT 5.1 post Sharpe ratios close to QMI (0.70 and 0.73 vs 0.74) largely because their low volatility reflects fewer regime switches, but their much lower Returns (3.3% and 4.4% vs QMI's 7.2%) point to weaker regime inference rather than true risk-adjusted skill.

Figure 16: Agentic AI Dynamic Factor Rotation | Europe L/S



Source: J.P. Morgan European Quantitative Strategy

Figure 17: Agentic Multi-Model Dynamic Factor Rotation | Europe L/S



Source: J.P. Morgan European Quantitative Strategy

Similar results are seen on a Long/Short return basis, with broadly strong performance, dominated by the committees, yet still inferior to the QMI. Subtle differences include:

1. The trend in newer models outperforming older ones is less clear-cut. In fact, the top 3 individual models for returns were GPT 5.5, GPT 5.1, then GPT 5.4.
2. Both multi-model committees outperform individual models to a more noticeable extent alongside a Drawdown reduction component which was somewhat lagging Long-only.
3. The Cycle Investing strategy delivered the strongest returns +13.7% at a Sharpe of 1.39. Several of the agent variants exceed that Sharpe by running lower volatility and shallower drawdowns, which is encouraging. The forward-looking focus should be on robustness: the committees show very high kurtosis and positive skew, indicating that *out*-performance may be more concentrated in a small set of episodes vis-à-vis broad-based accuracy across cycles.

Long/short is where the agents look most competitive. Several models exceed the Cycle Investing Sharpe, driven by materially lower Volatility.

The 2-member committee delivers the cleanest downside profile among scalable signals, even if its Sharpe is not the top.

4 models predict Expansion for September, and 2 predict Slowdown.

Table 7: Agentic AI Dynamic Factor Rotation | Long / Short | MSCI Europe

	t-Stat	Ann. Ret	Ann. Vol	Sharpe	Hit-Rate	Skewness	Kurtosis	Max DD
GPT 5.5	7.27	11.3%	8.7%	1.30	69.2%	1.57	13.87	-10.9%
GPT 5.4	6.97	10.5%	8.5%	1.24	69.5%	1.70	15.71	-13.5%
GPT 5.1	9.71	11.0%	6.3%	1.76	72.1%	0.08	2.30	-10.2%
OPUS 4.8	6.62	10.3%	8.8%	1.17	68.5%	0.78	8.72	-17.6%
SONNET 4.6	8.19	9.7%	6.6%	1.47	70.0%	0.35	4.22	-13.5%
HAIKU 4.5	11.03	10.1%	5.0%	2.00	73.3%	-0.12	0.93	-6.8%
6 member	8.51	12.0%	7.8%	1.53	72.8%	2.16	20.52	-13.7%
2 member	7.83	12.4%	8.8%	1.41	71.0%	2.26	18.54	-8.8%
QMI Cycle Investing	7.76	13.7%	9.8%	1.39	69.2%	1.44	11.36	-16.2%

Source: J.P. Morgan European Quantitative Strategy

An Agent's Guide to the Month Ahead

The latest model output from the August 2026 data pack, including its regime prediction for September and the rationale behind it can be found in Table 8. The models are split between expansion and slowdown. This lack of uniformity is not random: it reflects a genuinely mixed backdrop, sitting at the boundary between “positive and improving” and “positive but decelerating”; in other words, a turning point, where regime classification is hardest and a committee framework is most useful.

The more constructive calls emphasize PMIs above 50, benign volatility, tight credit spreads, and supportive risk appetite; the more cautious calls focus on softening UK momentum, mixed short-term Eurozone PMI changes, sticky inflation, and policy that remains restrictive.

Table 8: Model Reasoning - August 2026 data (September predictions)

Model	Drivers	Rationale	Decision
GPT 5.5	• Risk backdrop is supportive: VIX calm and credit spreads tight • Risk conditions are supportive: VIX calm and credit spreads tight • Inflation remains above target and ECB policy has tightened over 3-12m • EZ and UK manufacturing PMIs are both above 50 with positive 36m z-scores	Europe's activity level is still positive, with PMIs above 50 and benign risk conditions. The main ambiguity is momentum: EZ has re-accelerated near term, while UK manufacturing is decelerating, keeping Slowdown risk material.	Expansion
GPT 5.4	• Eurozone and UK manufacturing PMIs are above 50, keeping activity in positive territory. • Eurozone manufacturing PMI is back above 50 and well up over 6 months, but its 3-month change has softened. • Eurozone and UK manufacturing PMIs are both above 50, with PMI breadth 3/3 above 50	Europe looks more like positive-but-decelerating than outright weak: PMIs remain in expansion, but recent UK and EZ momentum has cooled. Stable labour data and benign risk appetite argue against contraction, while sticky inflation and recent ECB tightening cap the odds of full expansion.	Slowdown
GPT 5.1	• EZ and UK manufacturing PMIs are modestly above 50, but UK momentum is negative and EZ is mixed on a 1-3m basis • Unemployment rates in EZ and UK are stable at low levels, indicating still-positive activity • Inflation is slightly above target in EZ/UK with uneven 3-12m trends and modest ongoing ECB tightening • Global risk assets show tight credit spreads and low implied volatility, consistent with decelerating but not stressed conditions • Eurozone and UK manufacturing PMIs are modestly above 50 with positive 6m momentum but mixed short-term direction, indicating activity slightly above trend but not uniformly accelerating.	European activity is slightly above trend but losing momentum, particularly in UK manufacturing, while labour markets remain firm and inflation modestly above target. Policy is mildly restrictive and global risk indicators are calm, pointing to a positive but decelerating macro/profit backdrop consistent with a Slowdown rather than outright Expansion or Contraction.	Slowdown
Opus 4.8	• Tight HY/HG credit spreads, calm VIX (14.8) • Tight HY/HG credit spreads and calm VIX (14.8) • UK Mfg PMI 51.9 but softening (-0.6 1m, -1.8 3m) • UK PMI softening (-0.6 1m, -1.8 3m) • UK PMI softening (-0.6 1m, -1.8 3m) tempers upside	European PMIs sit above 50 with EZ improving and supportive risk conditions (calm VIX, tight spreads), favoring Expansion. But UK PMI is decelerating and EZ 3m change is negative, keeping meaningful Slowdown weight.	Expansion
Sonnet 4.6	• VIX at 20th percentile and HY spreads at 11th percentile signal strong risk appetite • EZ and UK Mfg PMI both above 50 with positive 6m delta, though 3m momentum mixed • US ISM strongly expansionary, providing positive external demand backdrop	PMIs above 50 with benign credit and vol conditions point to Expansion as base case, but short-term PMI deceleration in UK and flat EZ 3m trend introduce Slowdown risk. ECB easing and tight spreads are supportive; no contraction signals present.	Expansion
Haiku 4.5	EZ unemployment stable, UK labour market resilient Inflation moderating in UK, elevated in EZ; policy rates held steady EZ/UK PMI above 50 but momentum flat to negative (1m/3m) European unemployment stable; labour market not deteriorating	European PMI remains expansionary but 3-month momentum is deteriorating. Labour markets are stable, credit conditions are tight (supportive), and policy is on hold. Combination of above-trend activity with decelerating growth signals a Slowdown-dominant regime, with material Expansion tail risk if momentum stabilizes.	Expansion

Source: J.P. Morgan European Quantitative Strategy

Table 9: Committees View - Sept '26

Committee	Decision	Probability	Confidence	Votes
2 Member	Expansion	0.49	0.63	2
6 Member	Expansion	0.43	0.65	4

Source: J.P. Morgan European Quantitative Strategy

Both committee approaches predict an Expansion for September, with the 2-Member assigning a higher probability but lower confidence.

4/6 Models focus on still-strong macro data; only 2 focus on the marginal signs of slippage - an apt reflection of how the market often misses regime shifts, which appear first at the margin before headlines look weak.

This split seems an apt depiction of the current market outlook: high uncertainty, but with consensus still leaning bullish. That optimism is understandable. Despite ongoing geopolitical risks, rate uncertainty - made more concerning by extreme fiscal deficits, and debate over the sustainability of AI-related capex which currently underpins growth, investors have taken comfort from the resilience of markets so far, with many anchoring their bullish view on the AI-led growth story.

There is clearly enough strong data to justify a constructive view. However, the Slowdown calls, namely from GPT 5.1 and GPT 5.4, align more closely with our most recent published QMI signal, which moved to Slowdown in the US and Contraction in Europe in June 2026 and has remained there since. Rationale here focuses more on the margin: PMIs are still in expansionary territory, but momentum is fading, while persistent inflation keeps policy restrictive. In our own work, we also see the marginal direction of travel in some indicators beginning to soften, raising the question of whether the strong rally of recent years is approaching a less favorable risk/reward point.

It is therefore interesting that four models appear to lean into the good-growth story, while only two place enough weight on the fading momentum to cap the odds of expansion. This is not to dismiss the constructive view - which is well grounded in data - but it does echo a broader point from our cycle work: **investors often focus on whether data are still strong, while regime shifts are more often signalled by whether those data are starting to improve or deteriorate.**

Conclusions

- **Tightly-scoped agents can classify regimes coherently.** The data suggests LLM agents can produce stable, economically sensible regime sequences when the task is narrowly defined.
- **Date anonymisation helps, but it is not a cure.** Removing the date, and then removing both date and raw levels, still yields coherent regime paths, which is reassuring evidence the models can reason from the macro pack rather than leaning only on historical fingerprints.
- **Regime names are not doing the work when data is present.** Swapping “Expansion/Slowdown/Contraction/Recovery” for neutral labels leaves the mapping broadly intact with macro inputs, implying the models are anchored on the data pack more than pre-learned associations with familiar cycle terms.
- **Look-ahead risk remains real.** When given only the date, models still produce plausible macro narratives, confirming residual historical knowledge, and meaning true out-of-sample reproducibility cannot be guaranteed.
- **Directional skew is a first-order issue.** Many models are structurally pessimistic, calling Contraction and Slowdown far more often; the skew is materially less severe in newer generations.
- **Committees help, but composition matters more than size.** Ensemble models can dilute single-model bias, but the 6-Member committee inherits the shared pessimistic skew; the selective 2-Member (GPT 5.5 + Opus 4.8) produces a more balanced regime mix that aligns more closely with the QMI.
- **Accuracy is meaningful, not cosmetic.** Here “accuracy” is defined as the share of regime calls matching the QMI; reassuringly, newer models are not just more accurate, their calls also translate into better realised returns.
- **The real test is not classification, it is implementation.** When used as live rotation signals, all models generate positive excess returns, and performance improves with model generation; the 2-member committee is more profitable than the broader committee that includes older, skewed models.
- **Bottom line:** the J.P. Morgan QMI remains the trusted reference, but the opportunity is to use Agentic AI as a supplementary input to systematic factor timing as model capabilities continue to improve.

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